

◆ Progress Tracker Document

1. Introduction

The Progress Tracker is a web-based application developed to manage and monitor the implementation of Core Banking Solution (CBS) projects for different banks. It helps project teams plan, track, and manage project activities from the beginning of the project until its completion.

The application organizes the project into phases, milestones, tasks, subtasks, and activities, making it easier to monitor progress and manage the implementation process. It also provides features such as project information management, Excel schedule upload, activity approvals, notifications, audit logs, milestone weightage management, and report generation.

The Progress Tracker replaces manual Excel-based project tracking with a centralized system that improves project visibility, collaboration, accuracy, and overall project management.

1.1 Purpose

The purpose of this Functional Design Document (FDD) is to describe the functional requirements and business processes of the Progress Tracker application.

This document explains how each feature of the application works, the business rules followed by the system, and how users interact with different modules. It serves as a reference for business users, project managers, developers, testers, and other stakeholders to ensure a common understanding of the application's functionality.

1.2 Scope

The Progress Tracker application provides a centralized platform for managing and monitoring CBS implementation projects.

The application allows users to:

- Create and manage project information.
- Upload project schedules using Excel.
- Manage phases, milestones, tasks, subtasks, and activities.
- Submit and approve activity update requests.
- Assign milestone weightages.
- Monitor project progress through dashboards.

- Receive notifications for important project activities.
- Maintain audit logs for system changes.
- Generate project reports.
- Track the overall status of project implementation.

This document focuses only on the functional behavior of the application. It does not include technical implementation details, database design, source code, deployment process, or infrastructure configuration.

2 Key System Features

The Progress Tracker application provides the following key business features:

- User Authentication and Role-Based Access Control
- Project Information Management
- Project Schedule Upload
- Automated Project Structure Creation
- Project Dashboard and Progress Monitoring
- Activity Management
- Activity Update Approval Workflow
- Bulk Approval and Rejection
- Rollback of Activity Updates
- Milestone Weightage Management
- Project Risk Monitoring
- Audit Logging
- Notification Management
- Report Generation
- User Management
- Project Assignment
- Search and Filtering
- Dashboard Analytics

3. User Roles & Functional Modules

3.1 User Roles

The Progress Tracker application provides role-based access to ensure that each user can access only the features required for their responsibilities.

3.1.1 Administrator

The Administrator has full access to the application. The Administrator manages users, project information, projects, milestone weightages, audit logs, authorization requests, reports, and application settings. The Administrator can also upload project schedules and approve or reject activity update requests.

Responsibilities

- Manage users
- Manage projects
- Create project information
- Upload project schedules
- Manage milestone weightages
- Approve or reject activity updates
- View audit logs
- Generate reports
- Configure application settings

3.1.2 Implementation User

The Implementation User is responsible for managing project implementation activities. This user can upload project schedules, monitor project progress, manage activities, generate reports, and participate in project execution. The Implementation User cannot manage users or system settings.

Responsibilities

- Upload project schedules
- View dashboard
- Manage project activities
- View milestones
- View tasks

- Monitor project progress
- Generate reports
- View project documents
- Monitor project issues and risks

3.1.3 Management User

The Management User is responsible for monitoring project performance. This user can view dashboards, reports, project documents, issues, and risks to monitor the overall status of projects. The Management User has read-only access to project information and cannot modify project data.

Responsibilities

- View dashboard
- Monitor project progress
- View reports
- View project documents
- Monitor issues
- Monitor risks
- Review project status

3.1.4 User

The User is responsible for updating the progress of assigned project activities. Users can view assigned projects, update activity details, add remarks, submit activity update requests for approval, monitor project progress, and generate reports for their assigned projects.

Responsibilities

- View assigned projects
- View dashboard
- View milestones
- View tasks
- Update activity details
- Add remarks to activities
- Submit activity update requests
- Track project progress
- Generate reports

3.2 Login

Purpose

The Login module allows authorized users to access the Progress Tracker application securely.

Description

Users enter their username and password to log in. The system verifies the credentials before allowing access.

After successful login, users can access the modules based on their assigned role.

If the username or password is incorrect, the system displays an appropriate error message.

Main Features

- Secure user login
- Role-based access
- Session management
- Logout functionality

Inputs

- Username
- Password

Outputs

- Dashboard displayed after successful login
- Error message for invalid credentials

Business Rules

- Only active users can log in.
- Username and password are mandatory.
- Users can access only the modules assigned to their role.
- Charts and graphs
- KPI summary
- Notifications

Exception Scenarios

- No project assigned to the user.
- Selected project is unavailable.
- Dashboard data is unavailable.

3.3 Dashboard

3.3.1 Purpose

The Dashboard provides users with a clear view of project progress and current implementation status. The information displayed depends on the user's role and access permissions.

The application provides two different dashboards to meet the needs of different user roles.

3.3.2 Dashboard Types

The Progress Tracker application provides two dashboard views based on the user's role and business requirements.

3.3.2.1 Portfolio Dashboard

The Portfolio Dashboard is available to the following roles:

- Administrator
- Implementation User
- Management User

This dashboard provides a high-level overview of all projects across the organization. It helps management monitor project performance, portfolio progress, project health, and implementation status.

The Portfolio Dashboard includes:

- Bank Filter
- Project Search
- Total Banks
- Active Projects
- Overall Portfolio Progress
- Go-Live Summary

- On Track Projects
- At Risk / Delayed Projects
- Project Cards
- Project Readiness
- Current Phase
- Project Health
- Project Status

Users can select a project to view detailed project information.

3.3.2.2 Project Dashboard

The Project Dashboard is available to all user roles.

This dashboard provides detailed information about a selected project and helps users monitor project execution and progress.

The Project Dashboard includes:

- Project Overview
- Executive Summary
- Milestone Journey
- Activity Status Overview
- Risks & Issues Summary
- Overall Project Progress
- Activity Summary
- Milestone Status
- Project Notifications

The information displayed depends on the selected project and the permissions assigned to the logged-in user.

3.3.3 Functional Description

The Dashboard is displayed after a successful login.

Based on the user's role, the system displays either the Portfolio Dashboard or the Project Dashboard.

Portfolio Dashboard

1. The user logs in to the application.
2. The system displays the Portfolio Dashboard.
3. The user can filter projects by bank.

4. The user can search for a project.
5. The system displays portfolio statistics and project summaries.
6. The user can select a project.
7. By clicking **View Project Details**, the system opens the Project Dashboard for the selected project.

Project Dashboard

1. The user opens the Project Dashboard.
2. The system displays project information.
3. The user can select a milestone.
4. The user can apply date filters.
5. The system refreshes the dashboard based on the selected filters.
6. The user can monitor project progress, milestones, activities, risks, and issues.
7. Dashboard information is automatically updated whenever project data changes.

3.3.4 Business Rules

- Only authenticated users can access the Dashboard.
- Dashboard access depends on the user's role.
- Users can view only the projects assigned to them.
- Portfolio Dashboard is available only to Administrator, Implementation User, and Management User.
- Project Dashboard is available to all user roles.
- Dashboard information is refreshed whenever project data is updated.
- Filters display data only for the selected project, milestone, or date range.
- Dashboard displays the latest approved project information.

3.3.5 Inputs

The Dashboard accepts the following inputs:

Portfolio Dashboard

- Selected Bank
- Search Keyword
- Selected Project

Project Dashboard

- Selected Project

- Selected Milestone
- From Date
- To Date
- Time Filter (Today, Week, Month, Quarter, Half-Yearly, Year)

3.3.6 Outputs

The Dashboard displays:

Portfolio Dashboard

- Portfolio Summary
- Project Summary
- Project Cards
- Portfolio Progress
- Project Health
- Go-Live Information

Project Dashboard

- Project Overview
- Executive Summary
- Milestone Journey
- Activity Status
- Risks & Issues
- Project Progress
- Notifications

3.3.7 Exception Scenarios

The following situations may occur while accessing the Dashboard:

- No project is assigned to the user.
- No projects are available.
- No data is available for the selected filters.
- The selected project is inactive.
- The selected milestone does not contain any activities.
- The selected date range does not contain any project data.
- The user does not have permission to access the selected project.

3.4 User Management

3.4.1 Purpose

The User Management module allows the Administrator to manage application users and control their access to the Progress Tracker system.

It helps ensure that only authorized users can access the application and perform actions based on their assigned roles.

3.4.2 Overview

The User Management module is used to create, update, activate, deactivate, and delete users.

The Administrator can assign one or more projects to a user and define the user's role. Each role has different permissions within the application.

This module ensures secure access and proper user administration throughout the project lifecycle.

3.4.3 Functional Description

1. The Administrator opens the User Management module.
2. The system displays the list of registered users.
3. The Administrator can create a new user.
4. The Administrator can update user information.
5. The Administrator can assign one or more projects to a user.
6. The Administrator can change the user's role.
7. The Administrator can activate or deactivate a user account.
8. The Administrator can delete a user if it is no longer required.
9. The system saves the changes and updates the user list.

3.4.4 Features

The User Management module provides the following features:

- Create User
- View User List
- Search Users

- Update User Details
- Assign Projects
- Change User Role
- Activate or Deactivate User
- Delete User

3.4.5 Business Rules

- Only the Administrator can manage users.
- Username must be unique.
- Every user must have an assigned role.
- A user can be assigned to one or more projects.
- Inactive users cannot log in to the application.
- Deleted users cannot access the system.
- User information must be validated before saving.

3.4.6 Inputs

The following information is required while creating or updating a user:

- Full Name
- Username
- Password
- User Role
- Assigned Projects
- User Status

3.4.7 Outputs

The system provides the following outputs:

- User list
- User details
- Success message after create, update, or delete
- Validation message for invalid input
- Error message when an operation fails

3.4.8 Exception Scenarios

The following situations may occur:

- Username already exists.

3.5 Project Information

3.5.1 Purpose

The Project Information Feature is used to store the basic information required for a project before the implementation schedule is uploaded.

It ensures that all mandatory project details are available before project execution begins.

3.5.2 Overview

The Project Information Feature allows authorized users to create and manage project-related information.

The information entered in this module serves as the foundation for the project and is used during project implementation and reporting.

A project must have valid Project Information before the project schedule can be uploaded.

3.5.3 Functional Description

1. The user click on Create Project Button.
2. The user enters the required project details.
3. The system validates the entered information.
4. The system saves the project information.
5. The created project information becomes available for project implementation.

3.5.4 Information Captured

The Project Information module stores information such as:

- Bank / Client Information
- Product Information
- Project Details
- Project Manager Information
- Implementation Details
- Contact Information
- Go-Live Information

- Additional Business Information

3.5.5 Business Rules

- Only authorized users can create Project Information.
- Project Information must be created before uploading the project schedule.
- Each project should have only one Project Information record.
- Required fields must be completed before saving.
- Duplicate Project Information is not allowed.

3.5.6 Inputs

The user provides:

- Bank / Client Details
- Product Details
- Project Details
- Project Manager Details
- Go-Live Information
- Other required project information

3.5.7 Outputs

The system displays:

- Project Information
- Success message after saving
- Validation message for missing or invalid information

3.5.8 Exception Scenarios

- Required fields are missing.
- Project Information already exists.
- Invalid project details are entered.
- The user does not have permission to access the module.
- The system is unable to save the information.

3.6 Project Schedule Upload (Excel Upload)

3.6.1 Purpose

The Project Schedule Upload module allows authorized users to upload the project implementation schedule using the approved Excel template.

The uploaded Excel file is used to automatically create the complete project structure in the Progress Tracker application.

3.6.2 Overview

The Upload Project Excel module provides a simple interface for uploading the project schedule.

Users select the approved Excel template and upload it to the system. Before processing the file, the system validates the uploaded file and verifies the project information.

After successful validation, the system automatically creates the project hierarchy, including phases, milestones, tasks, subtasks, and activities.

3.6.3 Screen Description

The Upload Project Excel screen contains the following components:

- Upload Project Excel page
- Excel file selection area
- Supported file formats (.xlsx and .xls)
- Upload Excel button

The Upload Excel button remains disabled until a valid Excel file is selected.

3.6.4 Functional Description

1. The user opens the **Upload Project Excel** page.
2. The user clicks the upload area.
3. The user selects an Excel file.
4. The system validates the selected file.
5. The user clicks **Upload Excel**.
6. The system verifies the project information.
7. The system reads the Excel template.
8. The system creates the project structure.
9. The system displays a success message.

10. The uploaded project becomes available in the application.

3.6.5 Project Structure Creation

After a successful upload, the system automatically creates:

- Project
- Phases
- Milestones
- Tasks
- Subtasks
- Activities

The complete hierarchy is generated from the uploaded Excel template.

3.6.6 Business Rules

- Only authorized users can upload project schedules.
- Only **.xlsx** and **.xls** files are accepted.
- The uploaded file must follow the approved Excel template.
- Project Information must already exist before the upload.
- Duplicate project uploads are not allowed.
- The Upload button is enabled only after selecting a valid Excel file.
- The system creates the project hierarchy automatically after a successful upload.

3.6.7 Inputs

The user provides:

- Project Schedule Excel File (.xlsx or .xls)

3.6.8 Outputs

The system displays:

- Upload Success Message
- Validation Messages
- Error Messages (if any)

3.6.9 Exception Scenarios

The following situations may occur

- Project Information does not exist.

3.7 Milestone Weightage Management

3.7.1 Purpose

The Milestone Weightage Management module allows the Administrator to assign and update weightages for all milestones of a selected project.

The assigned weightages are used to measure project progress and implementation readiness.

3.7.2 Overview

After a project schedule is uploaded, the system automatically creates all project milestones.

The Administrator selects a bank and assigns a weightage to each milestone.

The total weightage of all milestones must always be **100%** before the changes can be saved.

This ensures that project progress is calculated consistently across all projects.

3.7.3 Screen Description

The Milestone Weightage Management screen contains the following components:

Bank Selection

The Administrator selects a bank from the available list. The system loads all milestones for the selected bank.

Milestone List

The system displays all milestones created from the uploaded project schedule.

For each milestone, the following information is displayed:

- Milestone Name
- Weightage (%)
- Current Status

Update Weightage

The Administrator can enter or modify the weightage for each milestone.

Total Weightage

The system automatically calculates the total of all milestone weightages and displays it at the bottom of the screen.

Update Weightage Button

After entering all weightages, the Administrator clicks **Update Weightage** to save the changes.

3.7.4 Functional Description

1. The Administrator opens the **Milestone Weightage Management** module.
2. The Administrator selects a bank.
3. The system displays all milestones for the selected project.
4. The Administrator enters or updates the weightage for each milestone.
5. The system automatically calculates the total weightage.
6. The Administrator clicks **Update Weightage**.
7. The system validates the entered values.
8. If the total weightage equals **100%**, the system saves the changes.
9. A success message is displayed.

3.7.5 Business Rules

- Only the Administrator can update milestone weightages.
- Every milestone must have a weightage.
- Weightage cannot be negative.
- Weightage must be a numeric value.
- The total weightage of all milestones must be exactly **100%**.
- The system does not save the data if the total weightage is not **100%**.
- The system automatically calculates the total weightage whenever a value is changed.

3.7.6 Inputs

The Administrator provides:

- Selected Bank
- Milestone Weightages

3.7.7 Outputs

The system displays:

- Milestone List
- Milestone Status
- Total Weightage
- Success Message
- Validation Messages

3.7.8 Exception Scenarios

The following situations may occur:

- Total weightage is not equal to **100%**.

3.8 Tasks

3.8.1 Purpose

The Tasks module allows authorized users to create, view, update, and monitor project tasks and activities throughout the project lifecycle.

It helps users track project execution, update task details, manage schedules, and record activity remarks.

3.8.2 Overview

The Tasks module is the primary working area of the Progress Tracker application.

Users can view all project tasks, apply filters, create new tasks, update existing task activities, add remarks, and generate task reports.

The module also provides a real-time summary of task progress and project execution.

3.8.3 Screen Description

The Tasks screen contains the following sections.

Project Information

Displays the selected project name.

Task Filters

The system provides the following filters:

- Phase
- Milestone
- Task
- Sub Task
- Activity
- Status
- Search

Task Summary

The system displays:

- Total Tasks
- Completed Tasks
- Delayed Tasks
- Not Started Tasks

Task List

The task table displays:

- Activity / Task Name
- Phase
- Milestone
- Start Date
- End Date
- Progress
- Status
- Actions
- Remarks

Available Actions

The following actions are available:

- Add Task
- Generate Report
- Edit Task
- Add / View Remarks

3.8.4 Create Task

The Create Task feature allows authorized users to add a new activity under an existing project hierarchy.

The user provides:

- Phase
- Milestone
- Task
- Sub Task
- Activity
- Owner
- Planned Start Date
- Planned End Date
- Actual Start Date
- Actual End Date
- Estimated Weeks

After validation, the system creates the task successfully.

3.8.5 Update Task

The Update Task feature allows users to modify task information.

Users can update:

- Activity
- Owner
- Planned Dates
- Actual Dates
- Estimated Weeks
- Progress
- Status

After successful validation, the updated information is saved.

3.8.6 Task Remarks

Users can add remarks for each task to record implementation updates.

The system allows users to:

- View previous remarks

- Add a new remark
- Save the latest remark

All remarks become part of the task history.

3.8.7 Functional Description

1. Open the **Tasks** module.
2. View the task list.
3. Filter tasks if required.
4. Create a new task or select an existing task.
5. Update task information.
6. Add remarks if required.
7. Save the changes.
8. The system updates the task information.

3.8.8 Business Rules

- Only authorized users can create tasks.
- Every task must belong to a Phase, Milestone, Task, and Sub Task.
- Required fields must be completed before saving.
- Progress must be between **0% and 100%**.
- Activity remarks cannot be empty.
- Updated task information is reflected in the project dashboard.

3.8.9 Inputs

- Task Details
- Owner
- Planned Dates
- Actual Dates
- Estimated Weeks
- Progress
- Status
- Remarks

3.8.10 Outputs

The system displays:

- Success Message

3.8.11 Exception Scenarios

- Required fields are missing.
- Invalid progress value.
- Invalid date values.

3.9 Activity Update Approval

3.9.1 Purpose

The Activity Update Approval module allows the Administrator to review, approve, reject, and revert activity update requests submitted by users.

It ensures that all project activity changes are reviewed before they become part of the project records.

3.9.2 Overview

Whenever a user updates an activity, the system creates an authorization request.

The Administrator reviews the requested changes, compares the previous and updated values, and decides whether to approve or reject the request.

The module also allows the Administrator to revert previously approved changes whenever required.

3.9.3 Screen Description

The Authorization module contains the following sections.

Summary Cards

The dashboard displays:

- Total Requests
- Pending Requests
- Approved Requests
- Rejected Requests

Search and Filters

The Administrator can search and filter authorization requests using:

- Search

- Request Type
- Request Status
- Requested User

Request List

The system displays:

- Request Type
- Requested By
- Activity Name
- Requested Date and Time
- Approval Status
- View Details

Bulk Actions

The Administrator can perform the following actions on multiple requests:

- Approve Selected
- Reject Selected

3.9.4 Activity Approval Details

When viewing a request, the system displays:

Request Information

- Project Name
- Activity Name
- Requested By
- Requested Date
- Approved By
- Approved Date

Change Summary

The system compares the previous and updated values.

The following information is displayed:

- Field Name

- Old Value
- New Value

Examples include:

- Actual Start Date
- Actual End Date
- Actual Period Week
- Progress
- Execution Status
- Schedule Health

3.9.5 Functional Description

1. The user updates an activity.
2. The system creates an authorization request.
3. The request appears in the Authorization module.
4. The Administrator reviews the request.
5. The Administrator compares the old and new values.
6. The Administrator approves or rejects the request.
7. The system updates the request status.
8. The action is recorded in the audit log.

3.9.6 Revert Changes

The system provides a **Revert Changes** option for approved requests.

When selected:

- The previous activity values are restored.
- The activity is updated.
- The rollback action is recorded in the audit log.

3.9.7 Business Rules

- Only the Administrator can approve or reject requests.
- Every activity update creates an authorization request.
- Users cannot approve their own requests.
- All approved and rejected requests are permanently recorded.
- The system maintains both old and new values for every modified field.
- Revert Changes is available only for approved requests.
- Every approval action is recorded in the audit log.

3.9.8 Inputs

The Administrator provides:

- Approval Decision
- Rejection Decision
- Bulk Approval Selection
- Bulk Rejection Selection
- Revert Request

3.9.9 Outputs

The system displays:

- Request List
- Request Status
- Change Summary
- Approval History
- Success Messages
- Validation Messages

3.9.10 Exception Scenarios

The following situations may occur:

- Request not found.
- Unauthorized user attempts approval.
- Activity information is unavailable.
- Invalid approval request.
- Revert operation fails.
- System error while processing the request.

3.10 Audit Log

3.10.1 Purpose

The Audit Log module provides a comprehensive history of all significant business operations performed within the Progress Tracker system. It ensures accountability, transparency, and traceability by recording every important action performed by authorized users. The module assists administrators in monitoring system activities, tracking changes, investigating issues, and maintaining historical records for compliance and operational review.

3.10.2 Overview

The Audit Log module automatically captures and stores information whenever users perform critical business operations. Every audit record includes details about the performed action, the affected entity, the project, the user who initiated the action, the execution timestamp, and, where applicable, the previous and updated values of the modified data.

The module maintains a complete chronological history of activities and provides powerful search and filtering capabilities for efficient record retrieval.

3.10.3 Screen Description

The Audit Log screen contains the following sections:

Search and Filter Panel

The Administrator can filter audit records using one or more of the following criteria:

- Project Name
- Action Type
- Entity Type
- Modified By
- Date Range
- Search Keyword

Audit Log List

The audit history is displayed in a tabular format containing:

- Date & Time
- Project Name
- Entity Type
- Entity Name
- Action Type
- Modified By
- Previous Value
- Updated Value

Audit Detail View

When an audit record is selected, the complete audit information is displayed, including:

- Previous Data
- Updated Data
- User Information
- Modification Timestamp
- Complete Change Summary

3.10.4 Functional Description

1. A user performs a business operation within the application.
2. The system validates and successfully completes the requested operation.
3. The system automatically generates an audit record.
4. The audit information is permanently stored in the database.
5. The Administrator opens the Audit Log module.
6. The system retrieves and displays audit records in descending order of execution time.
7. The Administrator may search or filter audit records based on business requirements.
8. The Administrator may view complete details of any audit record.

3.10.5 Audited Operations

The system automatically records audit information for the following operations:

Project Management

- Create Project
- Update Project
- Delete Project
- Upload Project Schedule

Activity Management

- Create Activity
- Update Activity
- Delete Activity
- Rollback Activity

Activity Approval Management

- Approve Activity Update
- Reject Activity Update
- Bulk Approve Activity Requests
- Bulk Reject Activity Requests

Milestone Management

- Update Milestone Weightage

User Management

- Create User
- Update User
- Delete User
- Reset User Password
- Activate User
- Deactivate User

Project Information Management

- Create Project Information

- Update Project Information
- Delete Project Information

3.10.6 Business Rules

- Audit records are generated automatically after successful completion of business operations.
- Audit records cannot be modified or deleted by application users.
- Each audit record stores the user responsible for performing the action.
- Each audit record records the exact execution date and time.
- Update operations store both previous and updated values.
- Rollback operations are maintained as independent audit entries.
- Audit history is available only to authorized users.
- Audit records are displayed in reverse chronological order, with the most recent records appearing first.

3.10.7 Inputs

The system automatically captures the following information:

- Project Name
- Entity Type
- Entity Name
- Action Type
- Previous Data
- Updated Data
- Modified By
- Action Date & Time

No manual input is required from the user.

3.10.8 Outputs

The system provides:

- Complete Audit History
- Audit Record Details
- Previous and Updated Values
- User Activity Information
- Search Results
- Filtered Audit Records

3.10.9 Exception Scenarios

The following exception scenarios are handled by the system:

- No audit records are available.
- No records match the selected search criteria.
- The selected project contains no audit history.
- The requested audit record cannot be found.
- The user is not authorized to access audit history.
- A system error occurs while retrieving audit information.

3.11 Report Generation

3.11.1 Purpose

The Report Generation module enables users to generate project progress reports based on selected filtering criteria. It provides a structured view of project activities and their execution status, allowing project stakeholders to monitor implementation progress, review milestones, and support project decision-making.

The generated report can be exported in Microsoft Excel format for analysis, presentation, and record-keeping.

3.11.2 Overview

The Report Generation module allows users to generate reports by selecting various filtering parameters such as project, phase, milestone, task, subtask, activity, execution status, and planned date range.

The system retrieves the matching project data and generates an Excel report containing the filtered information.

Users may generate reports for:

- Entire Project
- Selected Phase
- One or More Milestones
- Specific Task
- Specific Subtask
- Specific Activity
- Execution Status
- Planned Date Range

3.11.3 Screen Description

The Report Generation screen consists of the following sections.

Filter Panel

Users can specify one or more of the following filters:

- Project
- Phase
- Milestone (Single or Multiple Selection)
- Task
- Subtask
- Activity
- Execution Status
- Planned Start Date
- Planned End Date

Generate Report

After selecting the required filters, the user clicks the **Generate Report** button.

The system validates the selected criteria and generates the report.

3.11.4 Functional Description

1. User opens the Report Generation module.
2. User selects a project.
3. User selects one or more optional filters.
4. User clicks **Generate Report**.
5. The system validates the selected filters.

6. The system retrieves matching project information.
7. The system generates an Excel report.
8. The generated report is downloaded successfully.

3.11.5 Business Rules

- Project selection is mandatory.
- All remaining filters are optional.
- If no Phase is selected, all phases are included.
- If no Milestone is selected, all milestones are included.
- Multiple milestones may be selected simultaneously.
- If no Task is selected, all tasks are included.
- If no Subtask is selected, all subtasks are included.
- If no Activity is selected, all activities are included.
- If no Execution Status is selected, all execution statuses are included.
- If no planned date range is provided, the complete project schedule is included.
- Only records matching the selected filters appear in the generated report.

3.11.6 Inputs

The module accepts the following inputs:

- Project
- Phase
- Milestone(s)
- Task
- Subtask
- Activity
- Execution Status
- Planned Start Date
- Planned End Date

3.11.7 Outputs

The system generates an Excel report containing:

- Project Information
- Phase Details
- Milestone Details
- Task Details

- Subtask Details
- Activity Details
- Planned Dates
- Actual Dates
- Progress
- Execution Status
- Schedule Health
- Remarks

3.11.8 Exception Scenarios

The following exception scenarios are handled by the system:

- Project is not selected.
- Selected project does not exist.
- No matching records are found.
- Invalid filter combination is selected.
- Error occurs during report generation.
- Report generation fails due to system error.

3.12 Notification Management

3.12.1 Purpose

The Notification Management module provides real-time alerts to users regarding important project activities and system events. It ensures that users are promptly informed about activity update requests, approval decisions, project updates, and other significant actions without requiring them to manually monitor each module.

3.12.2 Overview

The system automatically generates notifications whenever predefined business events occur. Notifications are displayed through the notification bell available in the application header. The module

maintains both unread and previously viewed notifications, enabling users to stay informed about project activities.

Users can review recent notifications, mark all notifications as read, or navigate to the complete notification history for detailed information.

3.12.3 Screen Description

The Notification Management screen consists of the following sections:

Notification Bell

- Displays the total number of unread notifications.
- Opens the notification panel when clicked.

Notification Panel

The notification panel displays:

- Total unread notification count.
- Notification title.
- Notification description.
- Notification timestamp.
- Unread notification indicator.

Mark All as Read

- Marks all unread notifications as read with a single action.
- Updates the unread notification count immediately.

View All

- Redirects the user to the complete notification history page.
- Displays all notifications generated for the user.

3.12.4 Functional Description

1. A business event occurs within the application.
2. The system automatically generates a notification.
3. The notification is associated with the appropriate user.
4. The unread notification count is updated.
5. The user clicks the notification bell.

6. The system displays the most recent notifications.
7. The user may review individual notifications.
8. The user may mark all notifications as read.
9. The user may navigate to the complete notification history by selecting **View All**.

3.12.5 Notification Events

The system generates notifications for the following events:

- Activity Update Request Submitted
- Activity Update Approved
- Activity Update Rejected
- Bulk Activity Approval Completed
- Bulk Activity Rejection Completed
- Project Schedule Uploaded Successfully
- Activity Rollback Completed

3.12.6 Business Rules

- Notifications are generated automatically by the system.
- Notifications are displayed only to authorized users.
- Unread notifications are indicated using a notification badge.
- Notifications are displayed in descending order of creation time.
- Selecting **Mark All as Read** changes the status of all unread notifications to read.
- Read notifications remain available for future reference through the notification history.
- Notification records cannot be modified by users.

3.12.7 Inputs

No manual input is required.

The system automatically captures:

- Notification Title
- Notification Description
- Related Project
- Recipient User
- Notification Status
- Notification Date & Time

3.12.8 Outputs

The module provides:

- Unread Notification Count
- Recent Notification List
- Complete Notification History
- Notification Details
- Read/Unread Status

3.12.9 Exception Scenarios

The following exception scenarios are handled by the system:

- No notifications are available.
- No unread notifications are available.
- The requested notification cannot be found.
- The user is not authorized to access the notification.
- A system error occurs while retrieving notification information.

3.13 Revert Changes

Purpose

The Revert Changes functionality enables authorized users to restore an activity to its previous state after an activity update request has been approved. This feature helps recover from incorrect updates while maintaining complete audit history.

Functional Description

1. The user opens the Authorization module.
2. The user views an approved activity update request.
3. The system displays the previous and updated values of the activity.
4. The user clicks the **Revert Changes** button.
5. The system validates the request.
6. The previous activity information is restored.
7. The project schedule is updated successfully.
8. The system creates a new audit log entry for the rollback operation.
9. A success message is displayed to the user.

Business Rules

- Only authorized users can perform rollback operations.
- Rollback is available only for approved activity update requests.
- The system restores all previous activity values.
- Every rollback operation is recorded in the Audit Log.
- Original audit records remain unchanged after rollback.
- Users can review the restored values after successful completion.

Inputs

- Selected Approved Request
- Rollback Confirmation

Outputs

- Activity restored to its previous values.
- Updated project schedule.
- Rollback confirmation message.
- Audit log entry recording the rollback operation.

Exception Scenarios

- Selected request cannot be found.
- Activity no longer exists.
- Project cannot be found.
- User is not authorized to perform rollback.
- System error occurs while restoring activity information.